Parsimonious Machine Learning Methods for Time Series Forecasting of Valley Fever Infection in Maricopa County, AZ

By

Tiffany Le

A Thesis Submitted to The W.A. Franke Honors College

In Partial Fulfillment of the bachelor's degree
With Honors in

Statistics and Data Science

THE UNIVERSITY OF ARIZONA

MAY 2026

Approved by:

Dr. Yue Niu
Department of Mathematics

Acknowledgments

I would like to thank Dr. Selena Niu for her thoughtful mentorship throughout this research process. I thank Dr. John Galgiani and Dr. Andrew Comrie for their guidance and expertise and for helping me begin a project that has shaped my academic interests and deepened my passion for public health research. I also appreciate the faculty and colleagues whose insight and encouragement strengthened this work. Finally, I am grateful to my family for their unwavering support, and I dedicate this thesis to my father, Cuong Le, whose medical journey motivates my commitment to work that contributes to a safer and healthier world.

Abstract

As a fungal infection reportable in only twenty-seven U.S. states with symptoms remarkably like those of community-acquired pneumonia (CAP), Valley Fever diagnosis poses a unique and critical public health challenge in the Southwestern United States. Time series forecasting of infections among CAP patients is of particular interest for supporting healthcare providers in making informed testing decisions and motivating the public to adopt protective health practices during periods of elevated infection risk. With fifty-six Arizona clinics, Banner Urgent Care Services (BUCS) is one of the largest urgent care networks in Arizona. Maricopa County clinics represent over 80% of these. We performed a time series analysis of positive Valley Fever serologic results among CAP patients at Maricopa County BUCS clinics reported between December 1, 2019, and December 31, 2024, and developed machine learning models for forecasting future infections. We extended prior machine learning approaches for Valley Fever time series forecasting with an emphasis on parsimony, computational efficiency, and reliance on publicly accessible data for practical future integration into a real-time online tool. We restricted the Long Short-Term Memory (LSTM) model to daily atmospheric predictors and introduced the Gradient Boosted Decision Tree (GBDT) model as a time efficient alternative using lagged feature engineering. The LSTM and GBDT models demonstrated comparable performance on the validation set, with test MSEs of 0.3505 and 0.3699, respectively, equivalent to an average error of approximately three-fifths of an infection per day. Future infection rates among CAP patients in Maricopa County can be predicted with only daily wind speed, temperature, precipitation, PM10, and PM2.5 data, which are publicly available and easily integrated into automated systems.

Contents

1 Introduction

1.1 Background

Valley Fever, also known as coccidioidomycosis (CM), is a fungal infection endemic to the Southwest United States, namely in Arizona and California. It's caused by inhaling Coccidioides posadasii and immitis spores in soil. People with severe illnesses and weakened immune systems are at higher risk of infection and may face worse clinical outcomes, making diagnosis and treatment critical for these populations. While CM is easy to treat with antifungal medications, it's difficult to identify in patients and symptoms can persist for a prolonged period when left untreated.

The symptoms of CM are remarkably like those of community acquired pneumonia (CAP) as both conditions are characterized by cough, fever, fatigue and other respiratory issues. Approximately 29% of diagnosed CAP cases in Tucson, AZ are related to CM, further complicating the identification of the infection [1]. In addition to CAP, if left untreated, infection can lead to severe complications such as erythema nodosum and lung cavities. Patients with undiagnosed CM often receive empiric antibiotic treatment for presumed CAP, undergo unnecessary diagnostic tests, and suffer from preventable progression of destructive lesions beyond the lungs. Because CM is not reportable in twenty-three states, clinicians often fail to identify and test for it in patients with pneumonia-like symptoms [2].

As a first point of contact for patients with respiratory concerns, positive tests reported by urgent care clinics represent a significant subsection of county-wide CM diagnoses. Monthly Banner Urgent Care Services (BUCS) CM diagnoses and the rest of Maricopa County case counts were strongly correlated from 2020 to 2023. The challenges of accurate testing and the invasive procedures involved often make healthcare providers hesitant to administer CM tests. In 2023, only 23% of BUCS patients diagnosed with pneumonia were tested for CM [2].

Developing a predictive model for CM infections in Maricopa County BUCS clinics based on local atmospheric conditions would increase seasonal awareness of the infection. The tool would support healthcare providers in making more informed testing decisions during periods of increased infection risk in this vulnerable region. Understanding this risk empowers the general public to adopt healthier personal practices during high-risk periods, giving people agency over their own health, and helping reduce exposure and potentially prevent infections when environmental conditions favor CM transmission.

1.2 Related Work

The "grow and blow" hypothesis posits that seasonal and precipitation patterns influence the growth and sporulation of Coccidioides [3]. Wet periods with high precipitation create favorable conditions for fungal growth, while dry periods encourage sporulation and facilitate the aerosolization of spores. Work by Tamerius and Comrie (2011) supporting this hypothesis in Pima and Maricopa County revealed temporal autocorrelation in exposure rates from September through July, which abruptly ends in August. Furthermore, precipitation from October through December is highly correlated with exposure rates in the primary exposure season (August to March) in Maricopa, highlighting that the seasonality of exposure rates is driven by precipitation patterns [4]. In constructing our models, we used time lags and smoothing techniques to account for the effects of this hypothesis.

Powerful methods developed by Jin et al. and Sarabia et al. forecasted future infection rates using environmental and social predictors using Long Short-Term Memory (LSTM) [5] and Graph Neural Network (GNN) [6] models. However, the social COVID-19 Stringency Index calculated by the Oxford Coronavirus Government Response Tracker (OxCGRT) project included in previous models is no longer updated daily [7]. Fewer predictors may also be preferred by the Occam's razor principle for simpler models. Long Short-Term Memory models have large time complexity that requires significant computational power to reduce. We propose the Gradient Boosted Decision Tree (GBDT) as a faster alternative to LSTM for time series forecasting of CM infection that requires less data to learn temporal patterns. In practical applications for clinicians, epidemiologists, and the general public, smaller, more accessible, and computationally efficient models may be preferable, even at the cost of a modest reduction in predictive precision.

2 Methods

2.1 Data Collection and Preprocessing

As of 2024, BUCS comprised of fifty-six urgent care clinics in Arizona, forty-five of which were in Maricopa County [2]. These clinics represent a substantial share of the county's urgent care infrastructure, providing a representative patient population for monitoring regional CM cases among patients diagnosed with CAP. We analyzed a time series dataset of 1,858 daily observations of CAP cases, CM cases, CM diagnostic tests among CAP patients, and CM-positive results among CAP patients at Maricopa County BUCS clinics, spanning December 1, 2019 through December 31, 2024. Almost all coccidioidal serologic tests requested at BUCS utilize enzyme immunoassay that detects IgM and IgG antibodies produced by the immune system in response to the infection [2]. Consistent with the case definition set forth by the Council of State and Territorial Epidemiologists, we consider a single positive test as a confirmed case of coccidioidomycosis [8].

Not all patients diagnosed with CAP (as determined by having an ICD10 code of J18 associated with the visit) are administered a CM test. Daily counts of positive CM cases among BUCS patients with CAP significantly underestimates the number of patients with CAP that would have received a positive CM test result had they been tested. To adjust for the downward bias, we scaled the total number of patients with CAP by the proportion of positive CM cases among CAP patients. The distribution of CAP patients is slightly right-skewed, but the distribution of estimated CAP patients with CM exhibits a more pronounced right-skew as shown in Figure 1. While CAP diagnoses tend to occur frequently, positive CM cases are expected to be rare among these patients.

Symptoms begin to develop seven to twenty-eight days after infection, making it difficult to determine the exact date of infection. The date of visitation to an urgent care clinic could occur up to a month from initial infection. We applied a smoothing 30-day moving average to reduce short-term noise and account for temporal misalignment. The 30-day moving average of estimated positive CM tests is highly variable, reaching highs in 2021, 2024, and 2025 as shown in Figure 2A.

We collected the corresponding 1,858 days of daily aggregate climate and air quality data from the National Oceanic and Atmospheric Administration's (NOAA) Climate Data Online portal for the National Climatic Data Center [9] and the Environmental Protection Agency's (EPA) Outdoor Air Quality Data download tool [10]. We selected three meteorological stations in the metro Phoenix region of the county that are geographically closest to the main cluster of BUCS clinics. These stations are located near Phoenix Airport, Phoenix Deer Valley Municipal Airport, and Arizona State University. Not all meteorological stations monitor multiple atmospheric conditions but these three stations monitored the largest breadth of climate variables associated with CM incidence including temperature, precipitation, and wind speed. We selected three EPA ambient air quality monitoring sites in Central Phoenix, West Phoenix,

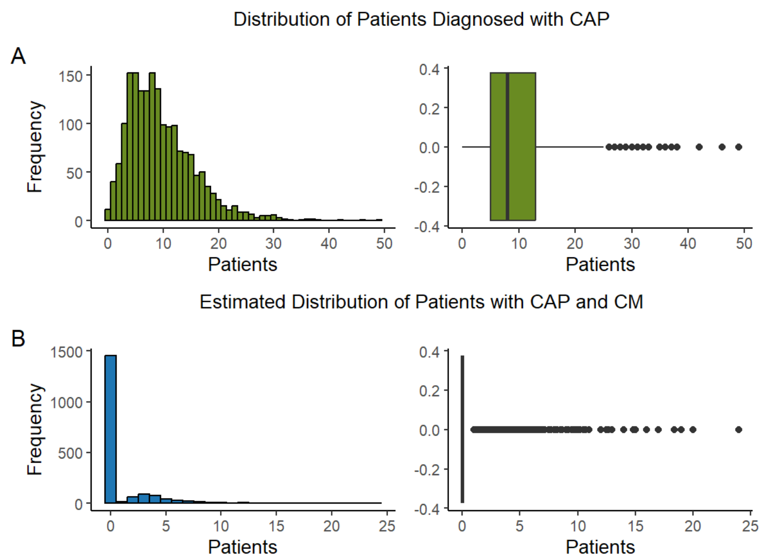

Figure 1: Distributions of BUCS patients diagnosed with CAP (A) and estimated patients diagnosed with CAP and CM (B).

and Tempe that are closest to the meteorological stations. These sites monitor concentrations of carbon monoxide, nitrogen dioxide, ozone, and sulfur dioxide, and record Air Quality Index, PM10 and PM2.5.

We aggregated daily atmospheric and air quality data with corresponding daily aggregate functions to constrain the data to a single time series for each variable and reduce the variation introduced by geographic location. We selected variables to include in machine learning models using best subset selection from preliminary linear regression models with Akaike Information Criterion (AIC). The AIC chose temperature (°F), PM10 ($\mu g/m^3$) and days from November 30, 2019 (the day before the first day in the time-series). We also included daily precipitation (in), daily wind speed (mph), and PM2.5 ($\mu g/m^3$) because they have documented associations with CM incidence [3].

Carbon monoxide was statistically significant at a 0.05 significance level in early linear regression models. The AIC criterion put great emphasis on this feature, but there is little scientific evidence that these levels should influence the incidence of CM so we decided not to proceed with the feature in following models. We hypothesize that carbon monoxide was acting as a proxy for an unmeasured social indicator as its levels often correlate with traffic volume which reflects urban activity patterns. For interpretability and reproducibility, we restrict our analysis to predictors with clear, measurable, and well-defined effects.

We applied the same 30-day moving average and normalized each variable to adjust for differences in scale and variance across predictors, allowing the models

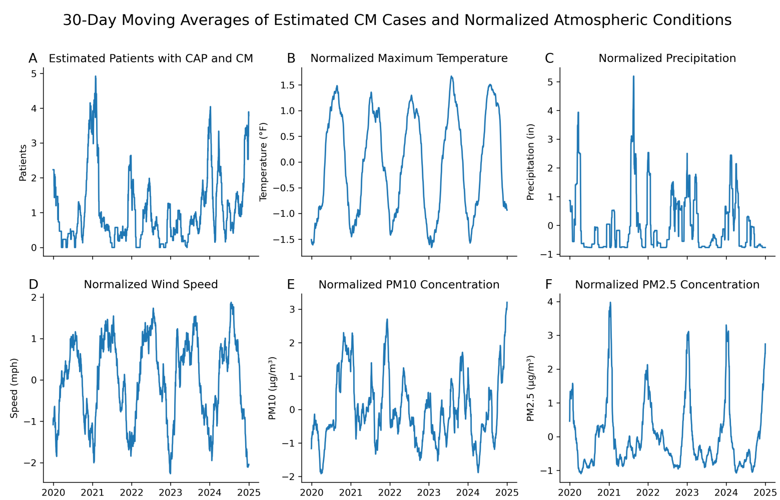

Figure 2: 30-day moving averages of estimated CM cases among CAP patients (A) and normalized atmospheric conditions including maximum temperature (B), precipitation (C), wind speed (D), PM10 concentration (E), and PM2.5 concentration (F).

to focus on temporal variation rather than on differences in magnitude driven by heterogeneous units or measurement ranges. Variables with the largest numeric scale often dominate sigmoid activations inside LSTM gates. Normalizing atmospheric variables prevents the domination of input gates by variables with the largest numeric scale. GBDT is invariant to monotonic transformations of predictors, so we apply the same scaling to this model to maintain consistency for comparison. The normalized 30-day moving average of PM10 (Figure 2E) closely follows the temporal pattern of the 30-day moving average of estimated CM cases (Figure 2A).

2.2 LSTM Model Training

We fit an LSTM model to the normalized and smoothed atmospheric data to forecast the smoothed time series of estimated positive CM tests using the Keras library [11]. We used the framework introduced by Jin et al [5] as a baseline. We iterated through many discrete values of each hyperparameter and chose values of each that minimized the validation mean square error (MSE). We settled on a thirty-day forecast window and dropout rate of 0.1. A full list of selected hyperparameter values is presented in Table A.1. While one of the optimal models presented by Jin et al. [5] has a batch size of five, a batch size of thirty-two minimizes our validation MSE and results in a faster training time than lower batch sizes.

While the validation loss far exceeded the training loss in initial passes through the training set during model optimization, they converged at approximately 200 epochs (Figure 3A). Loss convergence at a low value indicates effective generalization, indicating the model has captured the underlying signal rather than noise, and can consistently extend learned patterns to new data.

2.3 GBDT Model Training

We fit a GBDT model to the normalized and smoothed atmospheric data to forecast the smoothed time series of estimated positive CM tests. We include the same variables in the GBDT and LSTM models for an accurate performance comparison. We constructed a 120-day sliding window and flattened the past 120 days of every environmental variable into a single feature vector. The windowing process encoded all lagged information from 1 to 120 days, allowing the model to learn temporal dependencies directly from the sequential structure of the data.

Using defaults from the open-source eXtreme Gradient Boosting (XGBoost) library [12], we iterated through many discrete values of each hyperparameter and settled on a learning rate of 0.04, maximum tree depth of four, minimum child weight of three, no minimum loss-reduction threshold, 0.3 of data randomly sampled, and 0.7 features randomly sampled. A full list of selected hyperparameter values is presented in Table A.2. The model was trained over 300 boosting rounds, but training and validation MSEs began to converge at around 150 boosting rounds, indicating that additional trees no longer meaningfully reduced MSE (Figure 3B).

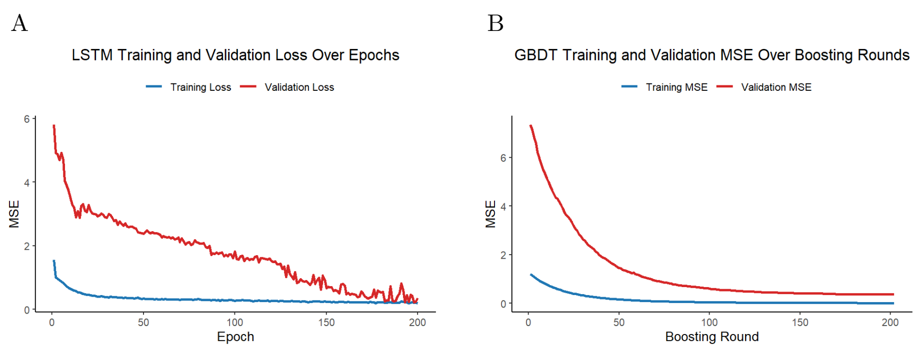

Figure 3: Decreasing and converging training and validation error over LSTM epochs (A) and GBDT boosting rounds (B).

3 Results

The LSTM and GBDT models demonstrated comparable performance on the validation set, with test MSEs of 0.3505 and 0.3699, respectively. The LSTM model predicted within 0.5920 of the actual estimated daily cases, compared to 0.6082 for the GBDT model. Time series of true and LSTM-predicted estimated cases show consistent agreement across both the training and validation periods, indicating effective generalization (Figure 4A). In contrast, true and GBDT-predicted cases align closely during training but exhibit larger deviations in the test period (Figure 4B).

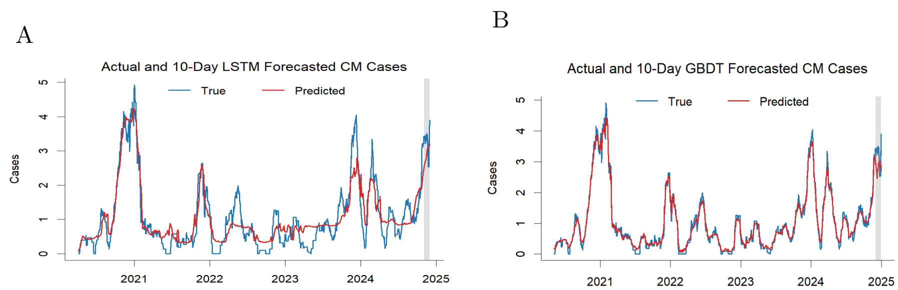

Figure 4: True and predicted time series of estimated CM cases for the LSTM (A) and GBDT (B) models with the validation set shaded gray.

We used Shapely Additive Explanations (SHAP) to determine how much each feature influences the magnitude of predictions. Analysis of absolute Shapely values showed that temporal position dominated feature importance in the LSTM model, indicating the model's strong reliance on historical sequence information. The mean absolute SHAP values of the day feature are double those of the environmental features (Figure 5). In contrast, the GBDT model distributed importance more evenly across predictors, indicating that it relied primarily on daily atmospheric and air quality variables rather than temporal ordering. The PM10 feature had the strongest influence on GBDT prediction while the temperature feature had the weakest influence. However, the range of mean absolute SHAP values is very small under the GBDT model. PM2.5 shows a small overall importance between the two models, despite PM10 having a high importance relative to the other environmental features, emphasizing the role of larger particulate matter, such as spores, in driving exposure rather than overall aerosol loading.

Computationally, GBDT prediction time remains near constant for fixed tree depth and number, while LSTM prediction time scales with sequence length for fixed number of hidden units and input dimension. Tree depth and the number of trees in the GBDT model, as well as the hidden units and input dimension

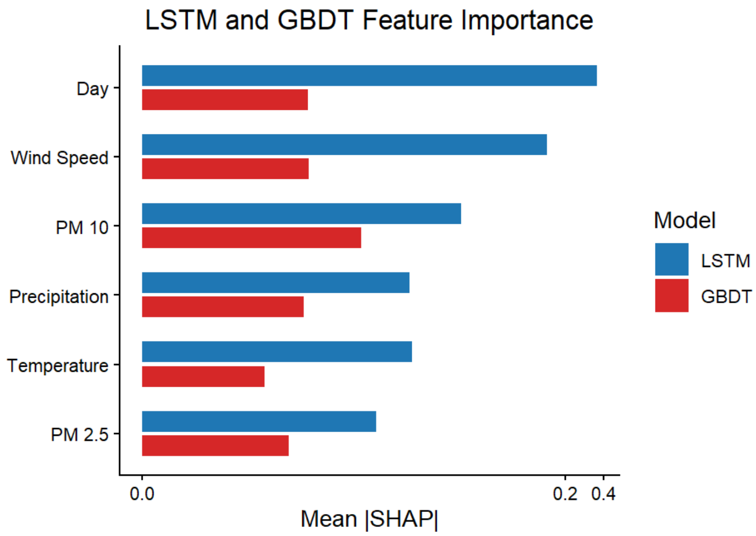

Figure 5: Mean absolute SHAP values for each feature in the LSTM and GBDT models. Features are presented in descending order based on overall feature importance between the two models.

in the LSTM model remain fixed because the same architecture is used for every prediction. The prediction time of the GBDT model stays approximately constant, while LSTM prediction time scales with the sequence length of the input. Thus, GBDT offers faster inference for long sequences with only a modest reduction in accuracy relative to the LSTM model, which generalizes slightly better on the thirty-day test set. Training and prediction complexities for both models are summarized in Table B.3

4 Discussion

Future infection rates among CAP patients in Maricopa County may be predicted with only daily wind speed, temperature, precipitation, PM10, and PM2.5 data which are publicly available and easily integrated into automated systems. Accurate forecasting does not require complex or scarce data. A small set of climatological and atmospheric features are sufficient for capturing environmental conditions associated with variation in CM infections.

The close alignment between true and GBDT-predicted cases during the training period and increased deviations during the test period suggest that the GBDT model overfits the training data slightly. The close training fit reflects the ability of the GBDT model to capture fine patterns in the training data, but these patterns do not always generalize to periods that aren't included in model training. Despite this, the test MSEs are similar for the LSTM and GBDT models, suggesting limited practical differences in predictive performance.

As a method typically used for regression and classification, it's natural that GBDT predictions were influenced largely by atmospheric and air quality data. The even distribution of feature importance under the GBDT model indicates the method may be more robust when little historical data is available. When new urgent care networks or regions begin time series forecasting future CM infections based on local conditions, it may take months before an LSTM model or GBDT model with long time lags accumulates enough data to produce stable forecasts. Although models with strong temporal dependence are more consistent across seasons, a GBDT model with shorter lags may be able to provide reasonable predictions earlier in deployment, offering a practical starting point for real-time forecasting systems.

The low prediction time of the GBDT model relative to the LSTM model for long sequences gives it a practical advantage for fast inference and when computational resources are limited. The prediction time of the GBDT model is effectively constant as structural parameters remain fixed. In contrast, LSTM model prediction time scales with sequence length, which becomes increasingly costly as longer historical windows are used. This computational distinction is important for operational forecasting systems that must generate predictions frequently or under constrained computational budgets.

Results suggest a trade-off between temporal generalization and computational efficiency. While the LSTM provides a slightly smoother representation of long-term temporal structure, the GBDT offers faster inference, earlier deployability, and comparable predictive accuracy. In research settings, LSTM models could achieve even higher accuracy by incorporating additional features, though this comes at the cost of increased model complexity and computational demand. From a clinical and public health perspective, however, the GBDT method is more accessible and efficient, making it practical for daily decision-making and routine monitoring.

5 Conclusion

Future infection rates among CAP patients in Maricopa County may be predicted within approximately three-fifths of the true number of estimated CM infections with only five environmental variables. These variables are publicly available and easy to incorporate in automated system. The LSTM and GBDT models presented could support a time-efficient, real-time online tool to aid healthcare providers in testing decisions and encourage the public to adopt healthier practices during periods of elevated Valley Fever infection risk.

This work suggests several pathways for future research and model development. A key next step involves automating the data querying and aggregation process, followed by developing front-end software for a daily forecasting dashboard or web-based application. Alternatives to the social COVID-19 Stringency Index like urban traffic patterns should be explored to reflect how much community behavior and mobility influence CM transmission. Early-warning thresholds may be developed to translate predictions into actionable alerts so that clinicians and public health departments wouldn't need to check predicted values daily.

Methods may be extended to urgent care clinics in other networks and counties. Outside the Southwest U.S., where CM is not endemic, infections are rare and more difficult to identify. Detecting CM in non-endemic regions is particularly challenging because low incidence reduces clinical awareness and diagnostic testing. The implementation of models into real-time online tools in endemic regions may increase awareness of the infection in non-endemic regions. As the tools become more widely adopted, clinicians and public health teams in other regions may become more familiar with CM trends, risk factors, and diagnostic cues, potentially improving recognition of cases that would otherwise be overlooked.

References

[1] L. Valdivia, D. Nix, M. Wright, E. Lindberg, T. Fagan, D. Lieberman, R. Cox, N. M. Ampel, and J. N. Galgiani, "Coccidioidomycosis as a common cause of community-acquired pneumonia," *Emerging Infectious Diseases*, vol. 12, no. 6, pp. 958–962, 2006.

[2] J. N. Galgiani, A. Lang, B. J. Howard, J. Pu, I. Ruberto, L. Koski, J. Collins, E. Rios, and T. Williamson, "Access to urgent care practices improves understanding and management of endemic coccidioidomycosis: Maricopa County, Arizona, 2018-2023," *The American Journal of Medicine*, vol. 137, no. 10, pp. 951–957, 2024.

[3] A. C. Comrie and M. F. Glueck, "Assessment of climate-coccidioidomycosis model: Model sensitivity for assessing climatologic effects on the risk of acquiring coccidioidomycosis," *Annals of the New York Academy of Sciences*, vol. 1111, pp. 83–95, 2007.

[4] J. D. Tamerius and A. C. Comrie, "Coccidioidomycosis incidence in Arizona predicted by seasonal precipitation," *PLoS ONE*, vol. 6, no. 6, 2011.

[5] X. Jin, F. Wei, S. S. Kandala, T. Umesh, K. Steele, J. N. Galgiani, and M. D. Laubichler, "Time series forecasting of Valley Fever infection in Maricopa County, Arizona using LSTM," *The Lancet Regional Health – Americas*, vol. 43, no. 101010, 2025.

[6] A. Sarabia, A. Sarabia, H. Yan, B. Sterner, and P. Jevtić, "Forecasting coccidioidomycosis (Valley Fever) in Arizona: A graph neural network approach," *arXiv*, 2025. [Online]. Available: https://arxiv.org/abs/2507.10014

[7] H. Ritchie, E. Mathieu, L. Rodés-Guirao, C. Appel, C. Giattino, E. Ortiz-Ospina, J. Hasell, B. Macdonald, D. Beltekian, and M. Roser, "Coronavirus pandemic (COVID-19)," https://ourworldindata.org/coronavirus, 2026.

[8] Council of State and Territorial Epidemiologists, "Update to the standardized surveillance case definition and national notification for coccidioidomycosis," Council of State and Territorial Epidemiologists, Atlanta, GA, Tech. Rep., 2022. [Online]. Available: https://cdn.ymaws.com/www.cste.org/resource/resmgr/ps/ps2022/22-ID-07_Coccidioidomycosis.pdf

[9] National Centers for Environmental Information, "Climate Data Online (CDO)," https://www.ncei.noaa.gov/cdo-web/, Asheville, NC, 2024, accessed July 2, 2025.

[10] U.S. Environmental Protection Agency, "Air data: Air quality data collected at outdoor monitors across the US," https://www.epa.gov/outdoor-air-quality-data, 2026, last updated January 29, 2026. Accessed July 2, 2025.

[11] J. J. Allaire, F. Chollet, RStudio, Google, Y. Li, J. Allaire, K. Ushey, Y. Tang, T. Kalinowski, J. Luraschi, E. Ruiz, H. Wickham, W. Chang, and R. François, *keras: R Interface to 'Keras'*, 2024, R package version 2.16.1. [Online]. Available: https://www.rdocumentation.org/packages/keras/versions/2.16.1

[12] T. Chen, T. He, M. Benesty, V. Khotilovich, Y. Tang, H. Cho, K. Chen, R. Mitchell, I. Cano, T. Zhou, M. Li, J. Xie, M. Lin, Y. Geng, Y. Li, J. Yuan, and D. Cortes, *XGBoost: eXtreme Gradient Boosting*, 2026, R package version 3.3.0.0. [Online]. Available: https://github.com/dmlc/xgboost

A Model Hyperparameters

This appendix provides the hyperparameters used in training for the LSTM and GBDT models. Reported values include those that influence model complexity, generalization, and training dynamics, while implementation defaults and non-modeling settings are omitted for clarity.

Table A.1: LSTM hyperparameters used in model training.

Hyperparameter	**Value**
Window Size	365
Horizon	30
Hidden Units	64
Activation Function	Tanh
Dropout Rate	0.1
Batch Size	32
Learning Rate	0.001
Optimizer	Adam
Epochs	200
Loss Function	Sigmoid

Dropout rate and batch size were chosen iteratively. Other hyperparameters were chosen in agreement with Jin et al [5] or to reduce training computational complexity.

Table A.2: GBDT hyperparameters used in model training.

Hyperparameter	**Value**
Objective	reg:squarederror
Evaluation Metric	RMSE
Learning Rate	0.03
Max Depth	4
Min Child Weight	3
Gamma	0
Subsample	0.3
Column Sample by Tree	0.7
Number of Boosting Rounds	300
Early Stopping Rounds	20

Learning rate, maximum tree depth, minimum child weight, loss-reduction threshold, proportion of data randomly sampled, and proportion of features randomly sampled were chosen iteratively.

B Complexity

This appendix summarizes the computational complexity of the LSTM and GBDT models. The expressions reported reflect worst-case prediction time complexities.

Table B.3: Computational complexity for model training and prediction.

Model	**Training Complexity**	**Prediction Complexity**
LSTM	$O\big(E\,T(h^2 + hd)\big)$	$O\big(T(h^2 + hd)\big)$
GBDT	$O(KN \log N)$	$O(KD)$

E denotes the number of training epochs, T the sequence length, h the number of hidden units, and d the input feature dimension. For the GBDT model, N represents the number of training samples, K the number of trees, and D the maximum tree depth.

C Scripts

Scripts used in data preprocessing, modeling, and visualization are available in the following GitHub repository:

`https://github.com/LeTiffany/Honors-Thesis.git`.